YEXIANG CHENG

School of Computer and Communication Sciences, EPFL
+041 767322858 ⋄ yexiang.cheng@epfl.ch ⋄ Linkedin

EDUCATION

Master in Computer Science, EPFL — Sep 2024 - Present

- GPA 5.44/6

Bachelor in Bioinformatics, Peking University — Sep 2020 - 2024

- GPA 3.69/4, Top 10% in School of Life Sciences

Bachelor in Computer Science (dual degree), Peking University — Sep 2020 - 2024

- GPA 3.73/4

RESEARCH EXPERIENCE

Research Assistant, AIMM, EPFL — Dec 2024 - Present
Advisor: Prof. Charlotte Bunne

- Downstream tasks of foundation models for spatial proteomics data, using Multiple Instance Learning framework to integrate embeddings for clinical label prediction and substructure identification.
- Constructed cellular level representation which enables cell retrieval with micro-environment information, improves zero-shot cell phenotyping, and supports biologically meaningful clustering that can automate biological discoveries.
- Vision Language Model for spatial proteomics data.

Research Intern, School of Life Sciences, Peking University — Sep 2023 - May 2024
Advisor: Lect. Fenglin Liu

- Conducted cell subgroup and gene expression analysis across Pan-Cancer datasets; developed classifiers to predict immunotherapy response and identified critical biomarkers using statistical methods, Shapley values, and LIME.

Research Intern, Wangxuan Institute of Computer Technology, Peking University — Nov 2022 - May 2023
Advisor: Prof. Jiaying Liu

- Developed a StyleGAN-based model for icon generation, incorporating separate heads to extract edge, texture, and color features. Designed a custom discriminator to enhance low-level feature learning.

Research Intern, Helixon — Jan 2022 - Jun 2022
Advisor: Prof. Jianzhu Ma

- Developed interpretable CNN models for macular degeneration classification, applying variational dropout to make the model focus primarily on the regions of interest to experts.

PUBLICATION

AI-powered virtual tissues from spatial proteomics for clinical diagnostics and biomedical discovery
J. Wenckstern*, E. Jain*, **Y. Cheng***, B. von Querfurth*, K. Vasilev, M. Pariset, P. F. Cheng, P. Liakopoulos, O. Michielin, A. Wicki, G. Gut, C. Bunne. *Preprint*

PROFESSIONAL EXPERIENCE

Teaching Assistant in Introduction to Computation (B) — Sep 2023 - Jan 2024

- Q&A support, assignment grading, and final exam design for C programming.

Teaching Assistant in Bioinformatics Lab — Feb 2023 - Jun 2023

- Assisted students with R and Linux; provided guidance on probability, statistics, and machine learning.

AWARDS AND SCHOLARSHIPS

- Excellent Graduate of Peking University — 2024
  (awarded to top 10% of students in Peking University)
- Excellent Graduate of Beijing — 2024
  (awarded to top 5% of students in Beijing)
- Qin Wanshun-Jin Yunhui Scholarship — 2023
  (awarded to top 10% of students in School of Life Sciences, Peking University)
- Merit Student — 2023
- UBIQUANT Scholarship — 2022
  (awarded to top 15% of students in School of Life Sciences, Peking University)
- Award for Academic Excellence — 2022
- The Third Prize of Peking University Scholarship — 2021
  (awarded to top 15% of students in School of Life Sciences, Peking University)
- Award for Contribution in Student Organizations — 2021
- Qin Wanshun-Jin Yunhui Scholarship — 2020
  (awarded to top 10% of students in School of Life Sciences, Peking University)
- Gold Medal in China National Biology Olympiad — 2019

SKILLS

Programming Languages	Python, C/C++, R, MATLAB
Technology & Frameworks	Pytorch, OpenCV, Transformers, Numpy, Pandas
Laboratory skills	Physiology, Cell Biology, Molecular Biology Techniques